

Yansen Zhang

City University of Hong Kong

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EDUCATION

2025.09–2025.12; **Visiting Student**, Machine Learning, Mohamed bin Zayed University of Artificial Intelligence
Supervisor: Prof. [Steve Liu](#)

2022.09–Present; **Ph.D. Candidate in Computer Science**, Computing, City University of Hong Kong
Supervisor: Dr. [Chen Ma](#)

Research Interests: Trustworthy Decisions; LLMs for OR; Data Valuation

2019.09–2022.06; **M.S. in Software Engineering**, Computer Science and Engineering, Sun Yat-sen University
GPA: 93.81/100 | Supervisor: Prof. [Yubao Liu](#)

Research Interests: Sequence-aware recommendation; Graph neural network

2015.09–2019.06; **B.S. in Software Engineering**, Software College, Northeastern University
GPA: 3.66/5 | Advisor: Prof. [Guibing Guo](#)
Postgraduate Candidates Exempt from Admission Exam

PROFESSIONAL EXPERIENCES

2022.04–Present; **Research on Trustworthy Recommendation**, City University of Hong Kong.

- Proposed a general data pruning framework by Shapley values in the recommendation. This work has been accepted by conference *KDD 2025*.
- Proposed a general framework of explainable diversity in the recommendation. This work has been accepted by conference *ECML PKDD 2025*.
- Presented a unified taxonomy of diversification metrics and approaches in retrieval systems from both the search and recommendation perspectives. This work has been accepted by journal *TKDE 2024*.

2019.09–2022.06; **Research on Graph-based Sequence-aware Recommendation**, Sun Yat-sen University.

- Focused on graph-based sequential recommendation mainly. Proposed a framework for sequential recommendation by a self-adaptive graph neural network with future contexts. This work has been accepted by conference *ICONIP 2021*.

2018.10–2019.06; **Research on Explainable Recommendation**, Northeastern University.

- Undergraduate Thesis: *Research and Implementation of Explainable Recommendation Based on Feature Fusion*. Achieved excellent score for the overall work.

RESEARCH INTERN EXPERIENCES

2026.02–2026.07; **Research on Self-Evolving Agent**, International Digital Economy Academy.

- Proposed an efficient framework for scientific discovery by leveraging LLMs. This work is intended to be submitted to conference *EMNLP 2026*.

2024.06–2025.08; **Research on LLMs for Operations Research**, Huawei Noah's Ark Lab.

- Proposed a unified framework of explainable operations research within large language models. In this work, we formulated the problem of explainable operations research within LLMs, and then we designed a unified framework to solve this problem, and also developed a benchmark to verify this issue. This work has been accepted to conference *ICLR 2025*.
- Proposed an iterative correction framework that improves the modeling ability in optimization. This work has been accepted to conference *ICML 2026*.
- Proposed a framework that compiles external knowledge into executable rules for forecast adjustment by LLMs and optimization methods. This work is intended to be submitted to conference *KDD 2027*.

CONFERENCE PUBLICATIONS

1. **Yansen Zhang**, Qingcan Kang, Yujie Chen, Yufei Wang, Xiongwei Han, Tao Zhong, Mingxuan Yuan, and Chen Ma (2026). *SAC-Opt: Semantic Anchors for Iterative Correction in Optimization Modeling*. In: *ICML*.

2. Ziqiang Cui, Yunpeng Weng, Xing Tang, Peiyang Liu, Shiwei Li, Bowei He, Jiamin Chen, **Yansen Zhang**, Xiquang He, and Chen Ma (2026). *CORE-RAG: Lossless Compression for Retrieval-Augmented LLMs via Reinforcement Learning*. In: ICML.
3. **Yansen Zhang**, Bowei He, Xiaokun Zhang, Haolun Wu, Zexu Sun, and Chen Ma (2025). *Counterfactual Multi-player Bandits for Explainable Recommendation Diversification*. In: ECML PKDD. [code].
4. **Yansen Zhang**, Qingcan Kang, Wing Yin Yu, Hailei Gong, Xiaojin Fu, Xiongwei Han, Tao Zhong, and Chen Ma (2025). *Decision Information Meets Large Language Models: The Future of Explainable Operations Research*. In: ICLR. [code].
5. **Yansen Zhang**, Xiaokun Zhang, Ziqiang Cui, and Chen Ma (2025). *Shapley Value-driven Data Pruning for Recommender Systems*. In: KDD. [code].
6. Chenhao Hu, Shuhua Huang, **Yansen Zhang**, and Yubao Liu (2022). *Learning to Infer User Implicit Preference in Conversational Recommendation*. In: SIGIR.
7. **Yansen Zhang**, Chenhao Hu, Genan Dai, Weiyang Kong, and Yubao Liu (2021). *Self-Adaptive Graph Neural Networks for Personalized Sequential Recommendation*. In: ICONIP. [code].

JOURNAL PUBLICATIONS

1. Shuliang Wang, Jiabao Zhu, Yi Wang, Chen Ma, Xin Zhao, **Yansen Zhang**, Ziqiang Yuan, and Sijie Ruan (2025). *Hierarchical Gating Network for Cross-Domain Sequential Recommendation*. TOIS.
2. Haolun Wu*, **Yansen Zhang***, Chen Ma, Fuyuan Lyu, Bowei He, Fernando Diaz, and Xue Liu (2024). *Result Diversification in Search and Recommendation: A Survey*. TKDE. [code].

ONGOING PUBLICATIONS

1. Mingyang Liu*, **Yansen Zhang***, Qingcan Kang, Xiaomin Deng, Xiongwei Han, Xiaojin Fu, Lei Chen, Tao Zhong, Mingxuan Yuan, Chen Ma, and Linqi Song (2026). "From External Information to Forecast: LLM-Based Time Series Forecasting with Constraint Modeling and Reflection Acceleration".

PATENTS

1. Yubao Liu, **Yansen Zhang** (2021). "A method for sequential recommendation model and sequential recommendation method." Chinese patent, 202111051076.3. Sun Yat-sen University.

INVITED TALKS

2025.08.26; **Explainable Recommendation via Bandit Optimization and Data Valuation** in Huawei Consumer Device Generative Recommendation Workshop, Huawei Hong Kong Research Center.

PROJECTS AND FUNDING

HK\$100,000; **Developing an LLM-RAG Based Course Assistant to Enhance Learning Experience**, funded by the City University of Hong Kong Learning: Experience, Achieve, and Develop (LEAD) Scheme.
 Student Team: Guanzhi Deng, Mingyang Liu, Jilin Cao, Ka Ki Yu, **Yansen Zhang**.
 Project Advisors: Prof. Linqi Song and Prof. Chen Ma.

TEACHING EXPERIENCES

Semester B 2025/26; **CS6487: Topics in Machine Learning**, City University of Hong Kong.
 Semester B 2022/23, 2023/24, 2024/25; **GE2324: The Art and Science of Data**, City University of Hong Kong.
 Semester A 2022/23, 2023/24, 2024/25; **CS2334: Data Structures for DS**, City University of Hong Kong.
 2019.10-2020.06; **Mathematical Analysis**, Sun Yat-sen University.

KEY SKILLS

- **Programming:** Python, C/C++, $\text{\LaTeX}$, Markdown, Matlab, SQL.
- **Frameworks/Platforms:** PyTorch, Linux, TensorFlow, Keras.

- **Software Tools:** Git, Visio, PhotoShop, MS Office.
- **Languages:** Chinese (native), English (IELTS 6.5).

SELECTED HONORS AND AWARDS

2024–2025;	Outstanding Academic Performance Award,	City University of Hong Kong
2021–2022;	Graduate Student Second–class Academic Scholarship,	Sun Yat-sen University
2019–2021;	Graduate Student First–class Academic Scholarship,	Sun Yat-sen University
2018;	Interdisciplinary Contest in Modeling (international-level),	Honorable Mention
2017–2018;	Excellent League Member Model (top 5%, departmental),	Northeastern University
2015–2017;	Excellent Students Awards (top 5%, departmental),	Northeastern University
2015–2018;	The Second Prize Scholarship,	Northeastern University
2015–2018;	National Encouragement Scholarship,	Northeastern University